Problem 1:

$$\dot{x}_1(t) = \frac{-1}{x_2^2(t)}$$

$$\dot{x}_2(t) = u(t)x_1(t)$$

Since $\tilde{u}(t) = 0 \quad \forall t \geq 0$,

$$\dot{\tilde{x}}_2(t) = 0 \quad \forall t \geq 0$$

$$\Rightarrow \tilde{x}_2(t) = C$$

since $\tilde{x}_2(0) = 1$, $C = 1$, so $\tilde{x}_2(t) = 1$

Hence $\dot{\tilde{x}}_1(t) = -\frac{1}{1^2} = -1$

$$\Rightarrow \tilde{x}_1(t) = -t + A$$

$\tilde{x}_1(0) = 1$, so $1 = 0 + A \Rightarrow A = 1$

Hence $\tilde{x}_1(t) = -t + 1 = 1 - t$

$$\dot{x} = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{x_2^2(t)} \\ u(t)\, x_1(t) \end{bmatrix} = f(x, u)$$

To linearize, compute the Jacobian

$$A(t) = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} \qquad -(x_2(t))^{-2}$$

$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 0 & \frac{2}{x_2^3(t)} \\ u(t) & 0 \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

plug in $\tilde{x}(t) = \begin{bmatrix} 1-t \\ 1 \end{bmatrix}$ and $\tilde{u}(t) = 0$



$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} 0 & \frac{2}{x_2^3(t)} \\ u(t) & 0 \end{bmatrix}_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 0 & \frac{2}{(1)^3} \\ 0 & 0 \end{bmatrix}$$

$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

$$B(t) = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} 0 \\ x_1(t) \end{bmatrix}_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}}$$

$$\tilde{x}(t) = \begin{bmatrix} 1-t \\ 1 \end{bmatrix} \quad \text{so} \quad B(t) = \begin{bmatrix} 0 \\ 1-t \end{bmatrix}$$

Now,

$$\delta\dot{x}(t) = A(t)\,\delta x(t) + B(t)\,\delta u(t)$$

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ 1-t \end{bmatrix} \delta u(t)$$

Problem 2:

State equations:

$$p(k+1) = p(k) + T_s\, v(k)$$

$$v(k+1) = v(k) + T_s\left(u(k) - c\,v(k)\,|v(k)|\right)$$

Output equation:

$$y(k) = v(k)$$

$$x(k) = \begin{pmatrix} p(k) \\ v(k) \end{pmatrix}$$

1) $x(k+1) = f(x(k), u(k))$

$y(k) = h(x(k), u(k))$

$$x(k+1) = \begin{pmatrix} p(k+1) \\ v(k+1) \end{pmatrix} = \begin{pmatrix} p(k) + T_s\, v(k) \\ v(k) - T_s\left(c\,v(k)\,|v(k)| + T_s u(k)\right) \end{pmatrix} = \begin{pmatrix} f_1(x(k), u(k)) \\ f_2(x(k), u(k)) \end{pmatrix}$$

$$= f(x(k), u(k))$$

$$y(k) = v(k) = h(x(k), u(k))$$

c) For Bruce Fine motion, el point

occur when

Given that $V^* > 0$ is constant

Define: $x = \begin{pmatrix} p(k) \\ v(k) \end{pmatrix}$

a) $v(k) = v^* > 0$ (constant)

Need $v(k+1) = v(k) = v^*$ to maintain velocity.

Since $v^* > 0$, $|v^*| = v^*$

$v(k+1) = v(k) - T_s c\, v(k)|v(k)| + T_s u(k)$

Substitute v^*

$v^* = v^* - T_s c\, v^{*2} + T_s u^*$

$T_s c\, v^{*2} = T_s u^*$

$$\boxed{u^* = c(v^*)^2}$$

b) since $y(k) = v(k)$, y evaluated at eq point yields:

$$\boxed{y^* = v^*}$$

c) nominal initial position: p_0^*

$$p^*(k) = p_0^* + kv^*$$

? need to multiply velocity by discrete timesteps since initial position. However, to account for sampling time, need to do kT_s.

thus,

$$\boxed{p^*(k) = p_0^* + kT_s v^*}$$

since $x_2 = v^*(k) = v^*$,

$$\boxed{x(k) = \begin{bmatrix} p(k) \\ v(k) \end{bmatrix} = \begin{bmatrix} p_0^* + kT_s v^* \\ v^* \end{bmatrix}}$$

3) Linearize around $(x^*(k), u^*)$

a) Define deviation variables:

$$\delta x(k) = x(k) - x^*(k)$$

$$\delta u(k) = u(k) - u^*$$

$$\delta y(k) = y(k) - y^*$$

b) Compute Jacobian Matrices:

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^*(k) \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial p} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial p} & \frac{\partial f_2}{\partial v} \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

$$= \begin{bmatrix} 1 & T_s \\ 0 & 1 - T_s c \frac{\partial}{\partial v}\left(v(k)|v(k)|\right) \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

$$x^*(k) = \begin{bmatrix} p_0^* + k T_s v^* \\ v^* \end{bmatrix}$$

$$= \begin{bmatrix} 1 & T_s \\ 0 & 1 - T_s c \frac{\partial}{\partial v}\left(v|v|\right) \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

Since $v^* > 0$, evaluate $\frac{\partial}{\partial v}\left(v|v|\right)$ as if $v > 0$.

Hence, $\frac{\partial}{\partial v}\left(v|v|\right) = \frac{\partial}{\partial v}\left(v^2\right) = 2v(k)$

$$A = \begin{bmatrix} 1 & T_s \\ 0 & 1 - 2cT_s v^* \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=x^*(k) \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

$$= \begin{bmatrix} 0 \\ T_s \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}} = \begin{bmatrix} 0 \\ T_s \end{bmatrix} = B$$

c)

$$C = \left.\frac{\partial h}{\partial x}\right|_{\substack{x=x^*(k)\\u=u^*}} = \left[\frac{\partial h}{\partial p} \quad \frac{\partial h}{\partial v}\right]_{\substack{x=x^*(k)\\u=u^*}}$$

$$= \begin{bmatrix} 0 & 1 \end{bmatrix}_{\substack{x=x^*(k)\\u=u^*}}$$

$$\boxed{C = \begin{bmatrix} 0 & 1 \end{bmatrix}}$$

$$D = \left.\frac{\partial h}{\partial u}\right|_{\substack{x=x^*(k)\\u=u^*}} = \left[\frac{\partial h}{\partial u}\right]_{\substack{x=x^*(k)\\u=u^*}}$$

$$\boxed{D = [0]}$$

d)

$$\delta x(k+1) = \begin{bmatrix} 1 & T_s \\ 0 & 1 - 2cT_s v^* \end{bmatrix} \delta x(k) + \begin{bmatrix} 0 \\ T_s \end{bmatrix} \delta u(k)$$

$$\delta y(k) = \begin{bmatrix} 0 & 1 \end{bmatrix} \delta x(k) + [0]\,\delta u(k)$$

4) $T_s = 0.1\,s$

$c = 0.2$

$v^* = 10 \frac{m}{s}$

$$A = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 - 2(0.2)(0.1)(10) \end{bmatrix} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 - 0.4 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.6 \end{bmatrix}}$$

$$\boxed{B = \begin{bmatrix} 0 \\ T_s \end{bmatrix} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix}}$$

$$\boxed{C = \begin{bmatrix} 0 & 1 \end{bmatrix}}$$

$$\boxed{D = [0]}$$

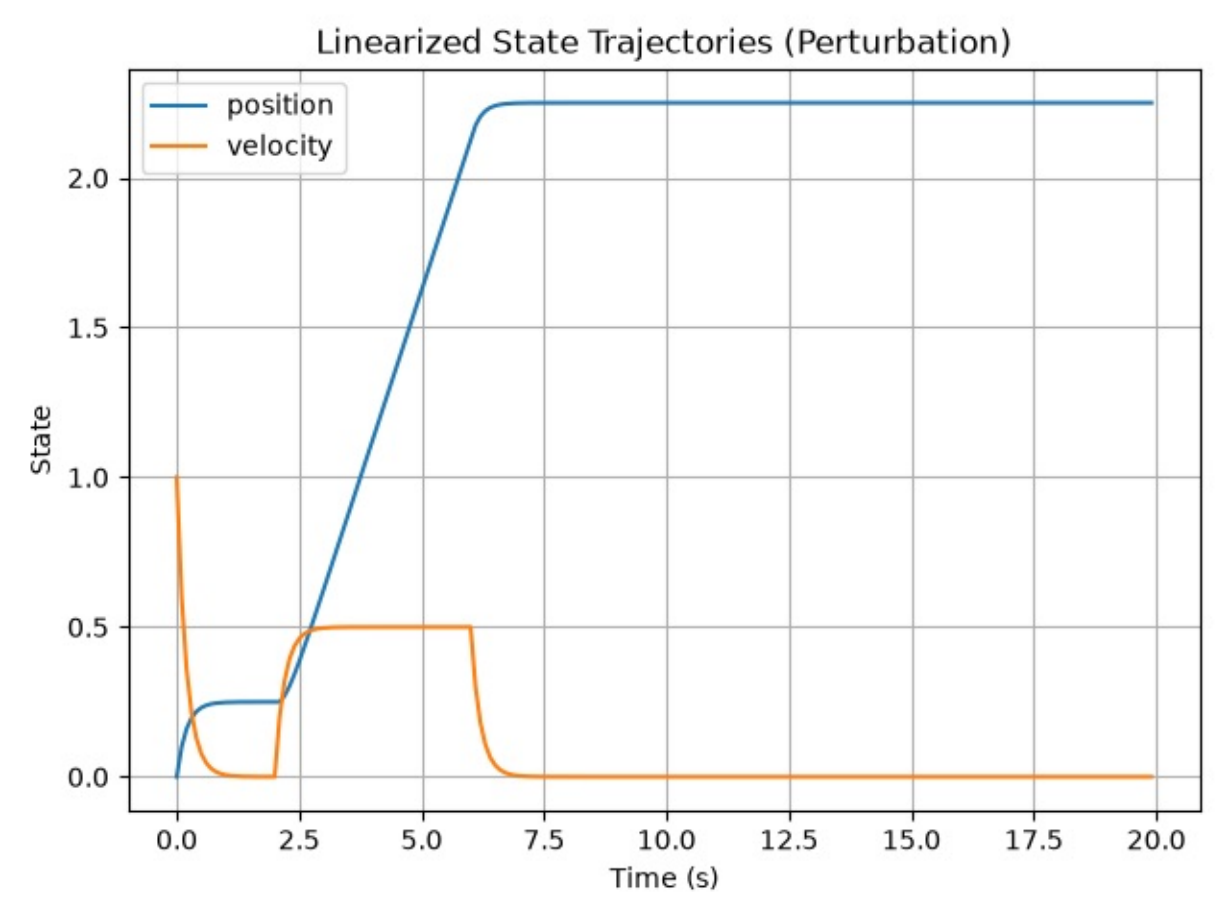
5)

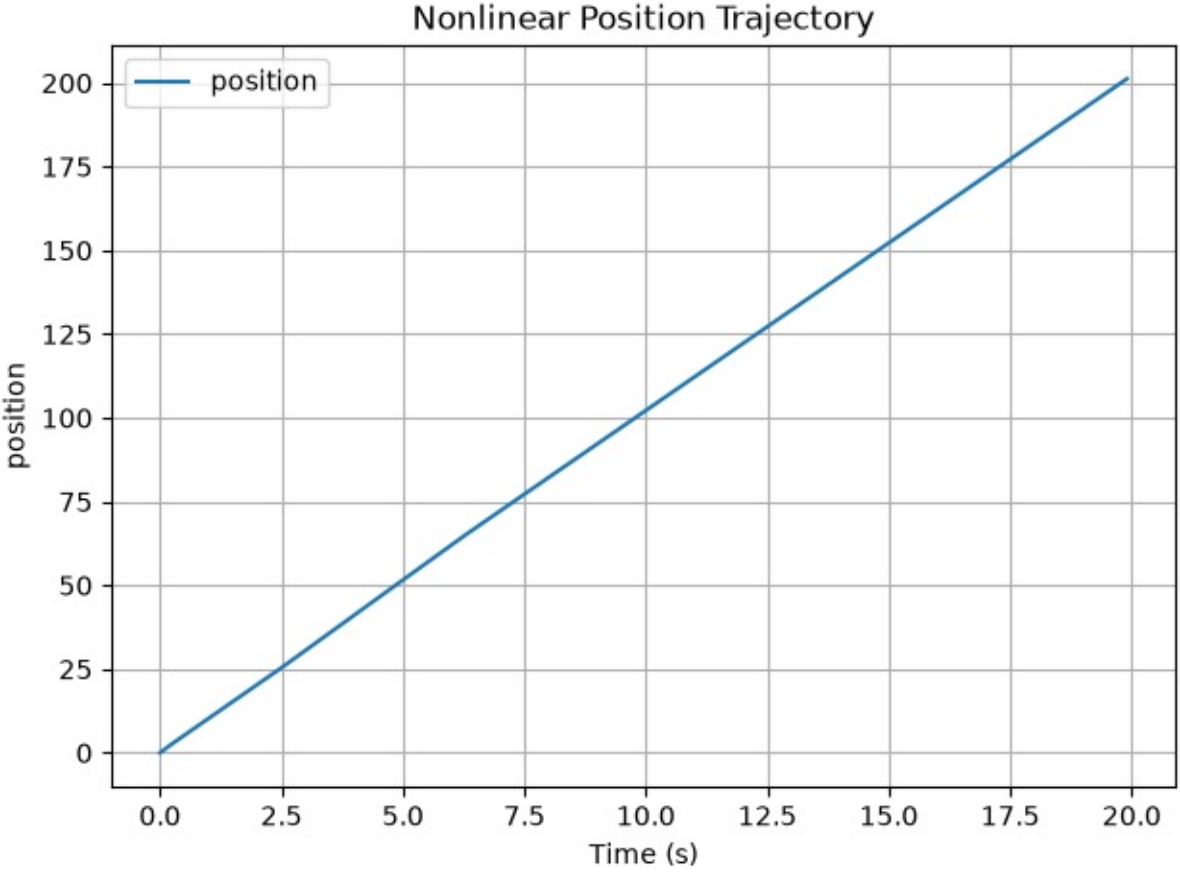
a) The linearized model depends on v^* because the input required to maintain a velocity of v^* changes as v^* increases. Additionally, v^* changes how fast the position $p(k)$ of the vehicle updates. Thus, the linearized model depends on v^*.

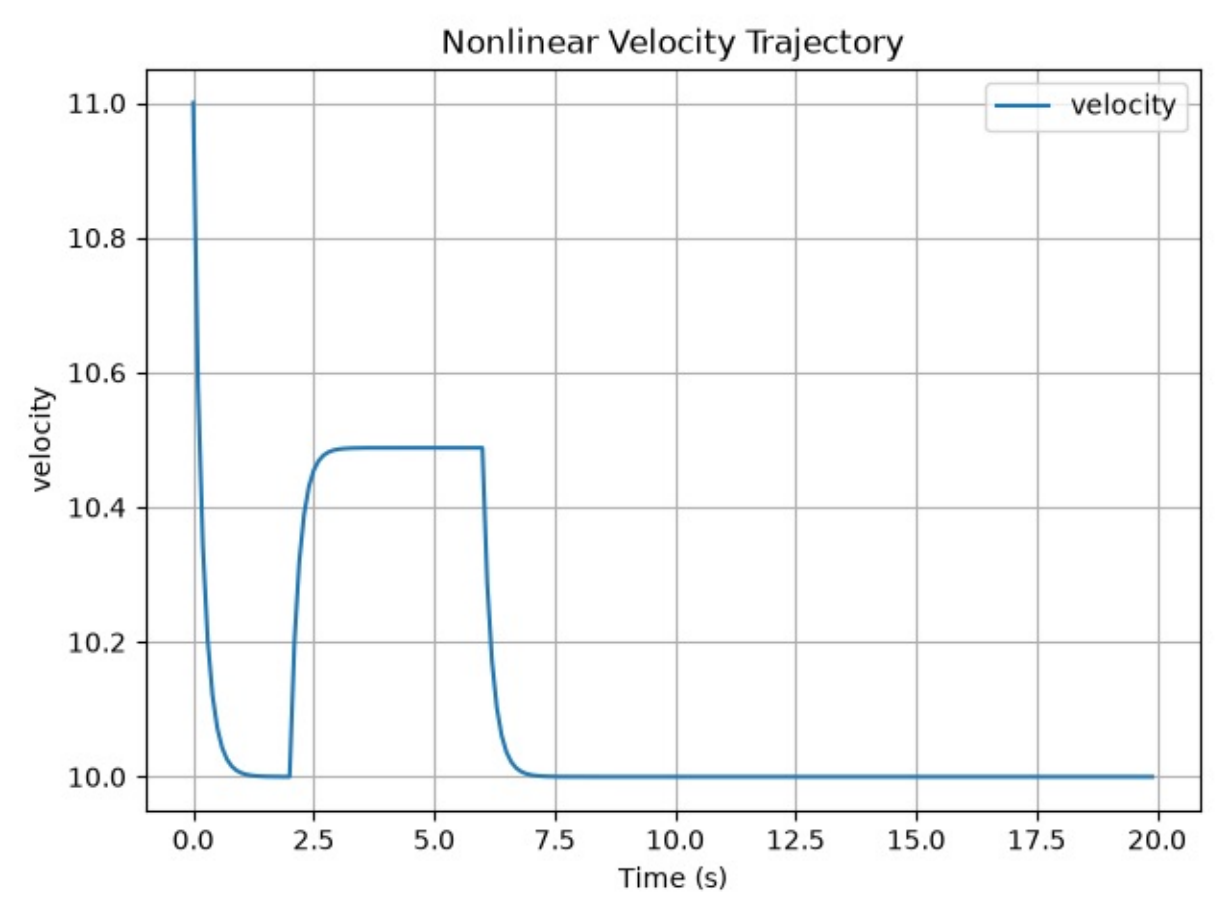
The linearized model also depends on v^* due to the $v(k)|v(k)|$ term in $v(k+1)$. The method of calculating $\frac{\partial}{\partial v}(v|v|)$ at v^* changes based on the sign of v^*.

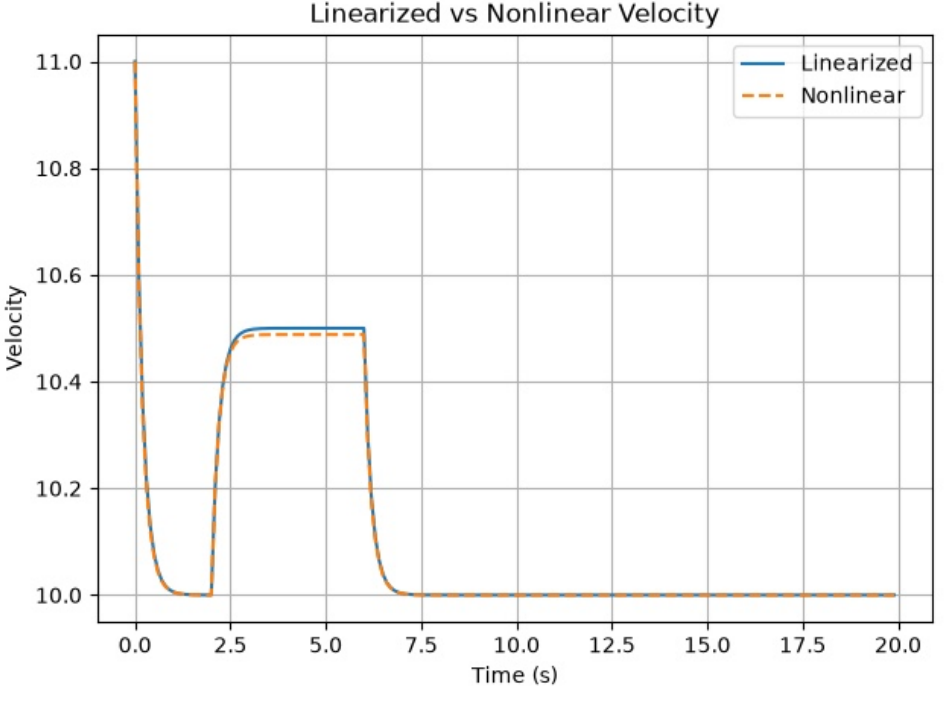
b) If the model was linearized at a different velocity, the only item that would change is the 2,2 entry of A.

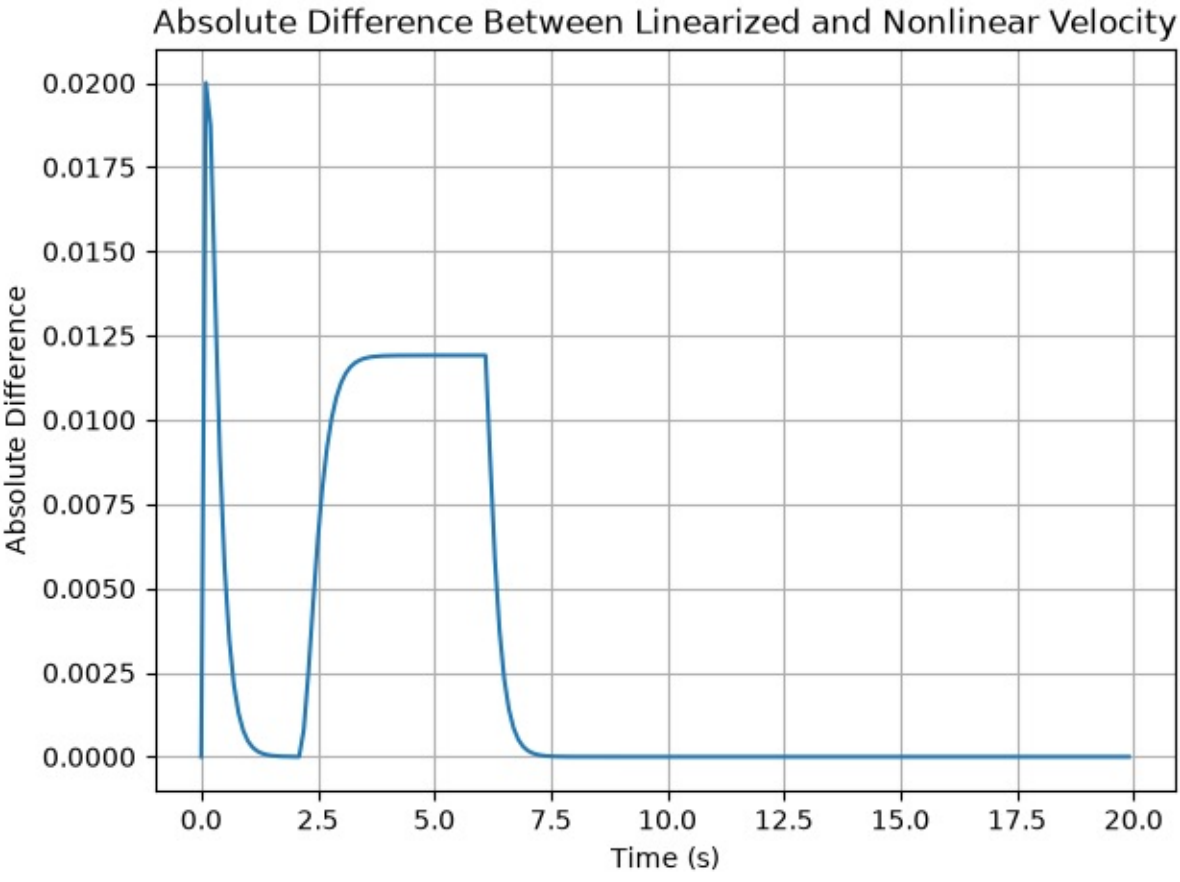
Additionally, the input required to maintain v^* would also increase.


Linearized State Trajectories (Perturbation)
position
velocity
State
Time (s)
0.0
0.5
1.0
1.5
2.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Nonlinear Position Trajectory
position
200
175
150
125
100
75
50
25
0
position
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Nonlinear Velocity Trajectory
velocity
11.0
10.8
10.6
10.4
10.2
10.0
velocity
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Linearized vs Nonlinear Velocity
Linearized
Nonlinear
Velocity
Time (s)
11.0
10.8
10.6
10.4
10.2
10.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Absolute Difference Between Linearized and Nonlinear Velocity
Absolute Difference
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

The output error is 0 when the perturbation is 0, for the most part. The slight increase in error at the beginning can be due to the nonlinear absolute value term in the nonlinear model. The zero error when perturbation is zero follows from the constraints of the linearized perturbation model, as it only aligns with the nonlinear model when the input perturbation causes the nominal trajectory to stay close to the nonlinear trajectory. However, When the perturbation (del(u(k)) is increased to 2 from k = 20 to 59, the output error increases. Evidently, this means that the perturbation of 2 was large enough to make the linearized model deviate from the nonlinear simulation.

Problem 2.3 Text:

Define $x_1(t) = \theta$, $x_2(t) = \dot{\theta}$, $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

We have:

$$ml^2\ddot{\theta} = mgl\sin\theta - b\dot{\theta} + T$$

$$T = \text{Sat}(u) = \begin{cases} -1 & u < -1 \\ u & -1 \le u \le 1 \\ 1 & u > 1 \end{cases}$$

$$y = \theta = x_1(t) = g(x, u)$$

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} \qquad \dot{x}_1(t) = \dot{\theta} = x_2(t)$$

$$\dot{x}_2(t) = \ddot{\theta} = \frac{mgl\sin\theta - b\dot{\theta} + T}{ml^2}$$

$$\dot{x}_2(t) = \frac{g}{l}\sin\theta - \frac{b}{ml^2}\dot{\theta} + \frac{T}{ml^2}$$

$$\dot{x}_2(t) = \frac{g}{l}\sin x_1(t) - \frac{b}{ml^2}x_2(t) + \frac{\text{Sat}(u)}{ml^2}$$

Thus,

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ \frac{g}{l}\sin x_1(t) - \frac{b}{ml^2} x_2(t) + \frac{sat(u)}{ml^2} \end{bmatrix} = f(x,u)$$

a) Given that eq. point exists where $\theta = x_1^* = 0$ Find x_2^* and u^*

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ \frac{g}{l}\sin x_1^* - \frac{b}{ml^2} x_2^* + \frac{sat(u^*)}{ml^2} \end{bmatrix}$$

Thus, $x_2^* = 0$.

$$\frac{g}{l}\sin \overset{=0}{x_1^*} - \frac{b}{ml^2}\overset{=0}{x_2^*} + \frac{sat(u^*)}{ml^2} = 0$$

$$\frac{sat(u^*)}{ml^2} = 0$$

$$sat(u^*) = 0$$

$sat(u^*) = 0$ when $u^* = 0$

now, $x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $u^* = 0$

Linearized Model:

$$\delta\dot{x}(t) = A\delta x(t) + B\delta u(t)$$

$$\delta y(t) = C\delta x(t) + D\delta u(t)$$

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

$$= \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(x_1) & -\frac{b}{ml^2} \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

Substitute $x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $u^* = 0$:

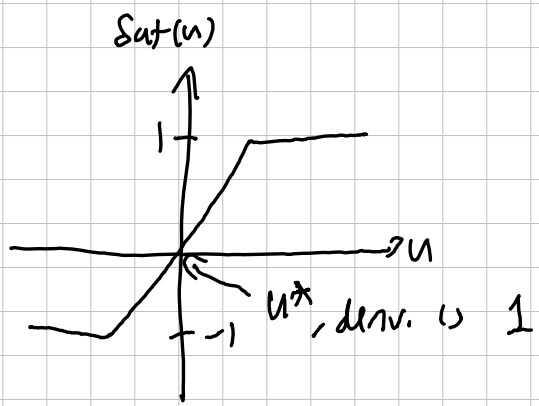
$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(0) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

$$= \begin{bmatrix} 0 \\ \frac{1}{me^2} \frac{\partial}{\partial u} sat(u) \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

Evaluate B at eq. point:

$$= \begin{bmatrix} 0 \\ \frac{1}{me^2} \frac{\partial}{\partial u} sat(u)\Big|_{u=u^*} \end{bmatrix}$$



$$B = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix}$$

$$g(x,t) = x_1(t) = \theta$$

$$C = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial u} \end{bmatrix} = [0]$$

Hence:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\, \delta u(t)$$

b) given that eq. point exists where $\theta = x_1^* = \pi$ Find x_2^* and u^*

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ \frac{g}{\ell} \sin x_1^* - \frac{b}{m\ell^2} x_2^* + \frac{sat(u^*)}{m\ell^2} \end{bmatrix}$$

$$x_2^* = 0$$

$$0 = \frac{g}{\ell} \underbrace{\sin(x_1^*)}_{\sin(\pi)=0} - \frac{b}{m\ell^2} x_2^* + \frac{sat(u^*)}{m\ell^2}$$

$$0 = \frac{sat(u^*)}{m\ell^2} \Rightarrow sat(u^*) = 0$$

$$\Rightarrow u^* = 0$$

Hence $x^* = \begin{bmatrix} \pi \\ 0 \end{bmatrix}$ $u^* = 0$

formulas for A, B, C, D stay same

but evaluate at new x^* and u^*

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(\pi) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \quad \text{(no substitution required)}$$

$$g(x,t) = x_1(t) = \theta$$

$$C = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial u} \end{bmatrix} = [0]$$

Hence:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\, \delta u(t)$$

c) given that eq. point exists where $\theta = x_1^* = \frac{\pi}{4}$ Find x_2^* and u^*

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ \frac{g}{l}\sin x_1^* - \frac{b}{ml^2}x_2^* + \frac{sat(u^*)}{ml^2} \end{bmatrix}$$

$$x_2^* = 0$$

$$0 = \frac{g}{l}\sin(x_1^*) - \frac{b}{ml^2}\overset{0}{x_2^*} + \frac{sat(u^*)}{ml^2}$$

$$0 = \frac{g}{l}\frac{\sqrt{2}}{2} + \frac{sat(u^*)}{ml^2}$$

$$-\frac{g\sqrt{2}}{2l} = \frac{sat(u^*)}{ml^2}$$

$$\frac{-mgl^2\sqrt{2}}{2l} = sat(u^*)$$

$$-\frac{mgl\sqrt{2}}{2} = sat(u^*)$$

range of sat(u) function is

$[-1, 1]$. Therefore, eq point only

exists when

$$-\frac{mgl\sqrt{2}}{2} \in [-1, 1], \text{ so } u^* \text{ can}$$

be solved for.

For rest of problem, assume that

$-\frac{mgl\sqrt{2}}{2} \in [-1, 1]$ so system can be

linearized. Further, you could say that

$-\frac{mgl\sqrt{2}}{2} \in [-1, 0]$ since $mgl \geq 0$ (nonnegative quantities realistically)

Hence

$$-\frac{mgl\sqrt{2}}{2} = u^* \quad \text{(assume to proceed)}$$

Formulation of A, B, C, D matrices stay the same, only substitution of x^* and u^* changes.

$$x^* = \begin{bmatrix} \frac{\pi}{4} \\ 0 \end{bmatrix} \qquad u^* = \frac{-mgl\sqrt{2}}{2}$$

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos\left(\frac{\pi}{4}\right) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{g\sqrt{2}}{2l} & -\frac{b}{ml^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \quad \text{(no switching required)}$$

$$y(t) = x_1(t) = \theta$$

$$C = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \left[\frac{\partial g}{\partial u}\right] = [0]$$

Final

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g\sqrt{2}}{2l} & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\,\delta u(t)$$

d) $b = \frac{1}{2}$ $\quad mgl = \frac{1}{4}$ compute

$T(t)$ needed so that pendulum follows

from $\theta(0) = 0$ w/ $\dot{\theta}(t) = 1 \quad \forall t \geq 0$

since $\dot{\theta}(t) = 1 \quad \forall t \geq 0, \quad \ddot{\theta}(t) = 0 \quad \forall t \geq 0$

$$ml^2\ddot{\theta} = mgl\sin\theta - b\dot{\theta} + T$$

$$0 = mgl\sin\theta - b + T$$

$$\tilde{T}(t) = b - mgl\sin\theta(t)$$

$$\tilde{T}(t) = \frac{1}{2} - \frac{1}{4}\sin\theta(t)$$

$$\dot{\tilde{x}}_1(t) = \dot{\tilde{\theta}}(t) = 1 \quad \forall t \geq 0$$

$$\tilde{x}_1(t) = t + C, \text{ since } x_1(0) = 0, \quad C = 0$$

$$\tilde{x}_1(t) = t$$

Hence, $\tilde{T}(t) = \frac{1}{2} - \frac{1}{4}\sin t$

$$\dot{\tilde{x}}_2(t) = \ddot{\tilde{\theta}}(t) = 0 \Rightarrow \tilde{x}_2(t) = \dot{\tilde{\theta}}(t) = 1 \quad \forall t \geq 0$$

Hence

$$\tilde{x}_1(t) = t$$

$$\tilde{x}_2(t) = 1$$

$$\tilde{T}(t) = \frac{s}{2} - \frac{1}{4}\sin t$$

Linearize around this trajectory.

The A, B, C, D matrix formula stay the same, but substitute time varying trajectories instead of equilibrium point ⟶

$$A(t) = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(x_1) & -\frac{b}{ml^2} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=0(t)}}$$

$$A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(t) & -\frac{b}{ml^2} \end{bmatrix}$$

$$B(t) = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \quad \text{(no saturation required)}$$

$$y(x,t) = x_1(t) = \theta$$

$$C(t) = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \left[\frac{\partial g}{\partial x_1} \quad \frac{\partial g}{\partial x_2}\right]_{\substack{x=\tilde{x}(t) \\ u=0(t)}} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D(t) = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \left[\frac{\partial g}{\partial u}\right]_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = [0]$$

Hence:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(t) & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\,\delta u(t)$$

Problem 2.4 Text

$$\ddot{\theta} + k\sin\theta = \tau$$

a) $u := \tau$ is input

$y := \theta$ is output.

Let $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$

Then $\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix}$

$\dot{x}_1 = \dot{\theta} = x_2$

$\dot{x}_2 = \ddot{\theta} = \tau - k\sin\theta = -k\sin x_1 + u$

$y = \theta = x_1$

Hence

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -k\sin x_1 + u \end{bmatrix} = f(x,u)$$

$$y = x_1 = g(x,u)$$

b) Find eq. points of system corresp to $z(t) = 0 \quad \forall t \geq 0$

At eq,

$f(x, u) = \vec{0}$. Hence:

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ -k \sin x_1^* + u^* \end{bmatrix}$$

$u^* = z^* = 0$ (given)

$x_2^* = 0$ by observation

$-k \sin x_1^* = 0$

$\sin x_1^* = 0$

$x_1^* = 0 + \pi k, \quad k \in \mathbb{Z}$

Hence:

$$x^* = \begin{bmatrix} 0 + \pi k \\ 0 \end{bmatrix}, \quad k \in \mathbb{Z}, \quad u^* = 0$$

c) Linearize system around solution

$$\tau(t) = \theta(t) = \dot{\theta}(t) = 0 \quad \forall t \geq 0$$

Treat given as trajectories:

$$\tilde{u}(t) = \tilde{\tau}(t) = 0$$

$$\tilde{x}_1(t) = \theta(t) = 0 \qquad \forall t \geq 0$$

$$\tilde{x}_2(t) = \dot{\theta}(t) = 0$$

Hence:

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 0 & 1 \\ -k\cos(x_1) & 0 \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$\tilde{x}(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \tilde{u}(t) = 0, \text{ so } A = \begin{bmatrix} 0 & 1 \\ -k\cos(0) & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -k & 0 \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \left.\frac{\partial y}{\partial x}\right|_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} & \frac{\partial y}{\partial x_2} \end{bmatrix}_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix}_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial y}{\partial u}\right|_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial y}{\partial u} \end{bmatrix}_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} 0 \end{bmatrix}_{\substack{x=\hat{x}(t) \\ u=\tilde{u}(t)}}$$

$$D = \begin{bmatrix} 0 \end{bmatrix}$$

Hence, linearized model is:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -k & 0 \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \end{bmatrix} \delta u(t)$$